

Pranav Balachander

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EDUCATION

The Pennsylvania State University
B.S. Computer Science | Dean's List

State College, PA
Graduated 2026

EXPERIENCE

Research Assistant

Dept. of Computer Science, Penn State

Harrisburg, PA

Apr. 2025 – May 2026

- Benchmarked 8 forecasting architectures across three concurrent research projects; the winning hybrid CNN-BiLSTM-Attention model in TensorFlow/Keras cut forecast error 25% (RMSE 12.40 vs 16.49) and raised R^2 to 0.87 from 0.63.
- Curated and standardized a 20-year day-ahead electricity price dataset across 40 markets and 5 continents, co-authoring the resulting paper accepted to IEEE Data Descriptions (2025).
- Built spatiotemporal ML pipelines in Pandas, NumPy, and Scikit-learn fusing quantum-chemistry, epidemiological, and genomic data sources.
- Applied SHAP across models to surface feature importance and make predictions interpretable and auditable.
- Authored three research papers as a named author spanning electricity price forecasting, Arctic mercury modeling, and pesticide-resistance surveillance.

AI/ML Intern

SBS IT Solutions

Dallas, TX

Jun. 2025 – Sep. 2025

- Contributed React front-end development to a HIPAA-compliant telehealth portal on a team of 5 to 8, helping design and build the main portal interface across doctor and patient views.
- Built the document ingestion path for the platform's RAG chatbot feature, parsing and chunking PDFs into a vector database served through LangChain and OpenAI APIs.
- Worked within HIPAA constraints on handling of patient messages and uploaded images; delivered to internal demo.

Data & Analytics Intern

mResult Corporation

Bangalore, India

Jun. 2023 – Oct. 2023

- Wrote SQL to clean, transform, and validate datasets of ~90,000 rows, producing reliable inputs for downstream analysis.
- Built and automated ETL flows in Dataiku, chaining prepare, join, and aggregation recipes over recurring data loads.
- Answered recurring business questions from management through exploratory analysis and visualizations in Python and Jupyter.

PUBLICATIONS

Unified Cross-Regional Day-Ahead Electricity Price Dataset (UniElecPrice)

IEEE Data

Descriptions, 2025

Co-authored a globally unified dataset of day-ahead electricity prices spanning 20 years, 40 markets, and 5 continents, enabling standardized cross-regional forecasting research. Benchmarked with a hybrid CNN-BiLSTM-Attention model that outperformed LSTM and Transformer baselines. Presented at the Harvard Undergraduate Research Conference, 2026.

PROJECTS

Commutr | React, Python, Figma

Apr. 2026 – Present

- Building a student commuting platform matching off-campus students with student drivers; led a team of 3 to 2nd place at the Harrisburg Launchbox Startup Pitch competition.
- Surveyed 200 students on commuting habits (72% commute 5+ days/week; 26% of walkable students skip class in bad weather) to segment users and scope core flows.
- Designed and built 12 screens covering .edu authentication, role selection, and separate driver and rider flows on a reusable component system.

- Built a multi-tenant AI workspace where an admin publishes prompts, skill files, HTTP function tools, and remote MCP servers to a whole team, while each member works in a private chat over shared organizational memory.
- Enforced per-server tool allowlists on both the provider side and locally, with a per-conversation approval gate that suspends a tool call across HTTP requests until the conversation owner approves it.
- Implemented an SSRF guard validating every DNS-resolved address against public IP ranges at both configuration and invocation time, with credentials stored as environment variable names rather than values.
- Wrote an audit layer persisting one row per tool call with arguments, result, and config revision, scrubbing secrets and bounding oversized responses before they re-enter the model context.

PetPilot | *TypeScript, Mastra, MCP, Arcade*

Winner, AI Valley Dogathon 2026

- Built a triage agent on top of PetPoint, the shelter system of record, that takes free-text staff and volunteer observations PetPoint has no field for, classifies them, and routes alerts to the responsible teams.
- Kept routing out of the model: a fixed lookup table maps each label to its teams, so every alert can be traced to the rule that sent it rather than to a generated decision.
- Implemented a confidence gate that blocks calendar booking and email drafting below threshold while still logging and alerting, so low-certainty events reach a human instead of triggering irreversible actions.
- Integrated Slack, Sheets, Calendar, and Gmail through an OAuth 2.1 MCP gateway, keeping user credentials out of the agent layer entirely.
- Built in a 5-hour window on a team of 4; in progress with the San Francisco SPCA to fit the router to their system.

Postcard | *Node.js, ClickHouse, PostgreSQL, MCP, React*

Better Days 2026

- Designed a dual-store memory system separating what the model can retrieve (ClickHouse, vector similarity over conversation history) from what it may assert (curated PostgreSQL rows), making grounding a structural property rather than a prompt instruction.
- Built an extraction pipeline that writes conversation-derived facts as unverified until a human confirms them, with deterministic guards and an audit trail of accepted and rejected extractions.
- Exposed a nine-tool MCP server backing the family console, and served the avatar through a custom LLM endpoint so retrieval runs inside the conversation loop.
- Ran embeddings in-process instead of over an API and moved scoring and extraction off the response path to hold conversational latency down.

JobSwipe | *MongoDB Atlas, Voyage AI, Flask, React*

MongoDB .local Build Fest

- Built a swipe-based job discovery app for new grads on a team of 3, gating postings on visa sponsorship and security clearance so users only see roles they are eligible for.
- Implemented semantic job matching with MongoDB Atlas Vector Search over Voyage AI embeddings.
- Served the matching pipeline through a Flask API to a React swipe interface.

Day-Ahead Electricity Price Forecasting | *Python, TensorFlow, Keras, SHAP*

Jul. 2025

- Built a hybrid CNN-BiLSTM-Multi-Head-Attention model forecasting 24-hour electricity prices from 48-hour history, owning the full training and evaluation pipeline.
- Cut forecast error 25% over LSTM and Transformer baselines across MAE, RMSE, SMAPE, and R^2 on real-world market data.
- Tuned LSTM units, dropout rate, and learning rate via Keras Tuner Bayesian optimization.
- Generated SHAP summary plots and prediction comparisons to surface feature importance and make predictions interpretable.

AI-Driven Modeling of Mercury Concentration in Arctic Environments | *Python, Scikit-learn*

Dec. 2025

- Built an ML framework fusing quantum-chemistry, molecular-dynamics, and environmental data to model mercury reduction, re-emission, and diffusion in Arctic snow and ice.
- Integrated quantum-chemistry calculations and molecular-dynamics simulations with data-driven models to capture multi-scale mercury behavior.
- Applied predictive modeling and validation across large simulation datasets to improve accuracy under changing conditions.

Assessing Tick-Borne Pathogen Transmission Risk in Pennsylvania | *Python, Scikit-learn, XGBoost*

Dec. 2025

- Benchmarked 4 ML models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost) on statewide PA tick surveillance data under 10-fold stratified cross-validation; Gradient Boosting won at 88.56% accuracy and 93.75% ROC-AUC.
- Engineered predictors from tick species, life stage, temperature, month, and geolocation, showing biological and temporal factors outweighed climate alone.
- Built county-level geospatial prevalence maps across Pennsylvania, identifying hotspots not explained by sampling effort.
- Work funded by an REU grant from Penn State Harrisburg's Office of Research and Outreach.

VALS: VR Anatomy Labeling Software | *Unity, C#, Capstone*

May 2026

- Led a team of 4 building a real-time label autocomplete system for a VR anatomy education app on Meta Quest 3, enabling in-headset search and assignment of anatomical labels to 3D structures.
- Co-implemented a Trie-based prefix search in C# supporting full-word and abbreviation matching, with the autocomplete UI built on TextMeshPro input fields.
- Loaded anatomy label datasets dynamically from CSV at runtime through Unity StreamingAssets, decoupling content from code.
- Shipped to the client with unit tests, pull-request code review, and written documentation.

Maargdarshak | *Python, Streamlit, OpenAI API*

HackPSU Fall 2024

- Built a documentation assistant answering natural-language questions about Penn State's ICDS user guide, on a team of 5 at HackPSU Fall 2024.
- Scraped and structured the ICDS documentation into a queryable source backing the OpenAI-powered answer pipeline.
- Designed and implemented the Streamlit front end, wiring it to the Python and OpenAI backend for real-time responses.

Online Restaurant Management System | *PHP, MySQL, HTML*

Sep. 2023

- Built a dynamic order-management platform backed by MySQL, architecting and indexing the schema for categories, orders, and customer records.
- Designed the staff-facing interface in PHP and HTML for managing food categories, orders, and customer records.
- Gathered user feedback and iterated features around automation and real-time updates.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, C#, PHP, HTML/CSS

ML & AI: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, SHAP, LangChain, RAG, OpenAI APIs, LLM agents, tool calling, MCP, structured outputs, embeddings, vector databases, prompt and context engineering

Data: Pandas, NumPy, ETL, data warehousing, Dataiku, Tableau, Power BI, Jupyter

Backend & Cloud: React, Node.js, FastAPI, Flask, Streamlit, REST APIs, AWS, Azure, Docker, Git, Linux, CI/CD

Databases: PostgreSQL, MySQL, SQLite, MongoDB, ClickHouse